

Engineering Mathematics

B.Tech 1st Year | 1st Semester

Unit II – Linear Differential Equations of Higher Order (Constant Coefficients)

Comprehensive Study Module | 40+ Fully Solved Problems | 50 Practice Exercises

Exam-Oriented | Step-by-Step Explained | Zero Steps Skipped

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1.1 What is a Differential Equation?

Definition: A **differential equation** is an equation that involves an unknown function and one or more of its derivatives. In engineering, differential equations model real physical systems such as electrical circuits, vibrating springs, heat flow, and wave propagation.

When engineers and scientists study how quantities change – for example, how current changes in a circuit over time, or how a spring oscillates – they write those relationships as differential equations. The unknown function (say y) depends on an independent variable (usually x or t), and the equation contains derivatives of y .

1.2 Why "Higher Order"?

A **first-order** differential equation involves only the first derivative dy/dx . When an equation involves second, third, or even higher derivatives, it is called a **higher order** differential equation. In engineering, most physical systems are governed by second-order or higher equations.

Example from Physics: Newton's second law $F = ma$, when applied to a spring-mass system with damping, gives:

$$m \cdot d^2x/dt^2 + c \cdot dx/dt + kx = F(t)$$

This is a **second-order linear ODE** – the backbone of vibration analysis, LCR circuits, and SHM.

1.3 General Form of a Higher Order Linear ODE with Constant Coefficients

The general n -th order linear ODE with constant coefficients is written as:

$$a_0 \cdot d^n y/dx^n + a_1 \cdot d^{(n-1)} y/dx^{(n-1)} + a_2 \cdot d^{(n-2)} y/dx^{(n-2)} + \dots + a_{(n-1)} \cdot dy/dx + a_n \cdot y = f(x)$$

where $a_0, a_1, a_2, \dots, a_n$ are **real constants** (not functions of x), and $f(x)$ is a given function of x called the **right-hand side** or **forcing function**.

1.4 The Operator Notation D

To simplify writing and solving, we introduce the **differential operator** D defined as:

$$D = d/dx \quad \text{so that} \quad D^2 y = d^2 y/dx^2 \quad D^3 y = d^3 y/dx^3 \quad \text{and so on.}$$

Using D -notation, the general equation becomes:

$$(a_0 \cdot D^n + a_1 \cdot D^{(n-1)} + \dots + a_{(n-1)} \cdot D + a_n) y = f(x)$$

We write this compactly as $f(D)y = f(x)$, where $f(D)$ is a polynomial in D .

Key Point: The coefficients $a_0, a_1, \dots, a_n$ must be **CONSTANTS** (numbers), not functions of x . When they are constants, powerful algebraic methods (auxiliary equation, operator methods) become available.

Section 2 Basic Definitions – Order, Degree, Linearity Must-Know Concepts

2.1 Order of a Differential Equation

Definition: The **order** of a differential equation is the order (highest power of index) of the **highest-order derivative** present in the equation.

Equation	Highest Derivative	Order
$dy/dx + y = x^2$	dy/dx (1st derivative)	1
$d^2y/dx^2 + 3 \cdot dy/dx + 2y = 0$	d^2y/dx^2 (2nd derivative)	2
$d^3y/dx^3 - 4 \cdot dy/dx = \sin x$	d^3y/dx^3 (3rd derivative)	3
$d^4y/dx^4 + y = e^x$	d^4y/dx^4 (4th derivative)	4

2.2 Degree of a Differential Equation

Definition: The **degree** of a differential equation is the power (exponent) of the highest-order derivative, after the equation has been made **free from radicals and fractions** involving the derivatives.

Equation	Highest Derivative & Its Power	Degree
$(d^2y/dx^2)^3 + dy/dx = x$	$(d^2y/dx^2)^3 \rightarrow \text{power} = 3$	3
$d^2y/dx^2 + (dy/dx)^5 = 0$	$d^2y/dx^2 \rightarrow \text{power} = 1$	1
$[1 + (dy/dx)^2]^{(3/2)} = d^2y/dx^2$	After squaring: $(d^2y/dx^2)^2 \rightarrow \text{power} = 2$	2

Important: When the equation involves terms like $\sin(dy/dx)$, $e^{(d^2y/dx^2)}$, or $\log(dy/dx)$, the degree is said to be **undefined (not defined)**, because the equation cannot be made into a polynomial in derivatives.

2.3 Linear vs Non-Linear Differential Equations

A differential equation is called **linear** if:

1. The dependent variable y and all its derivatives appear to the **first power** only (no y^2 , no $(dy/dx)^3$, etc.)
2. There are **no products** of y with any of its derivatives (no $y \cdot dy/dx$, etc.)
3. The **coefficients** of y and its derivatives are either constants or functions of x only

Equation	Linear / Non-Linear	Reason
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$d^2y/dx^2 + 5 \cdot dy/dx + 6y = 0$	LINEAR	y and derivatives appear to power 1; constant coefficients
$d^2y/dx^2 + y^2 = 0$	NON-LINEAR	y^2 term (y appears squared)
$d^2y/dx^2 + y \cdot dy/dx = x$	NON-LINEAR	Product of y and dy/dx
$x \cdot d^2y/dx^2 + 3y = \sin x$	LINEAR (but not const. coeff.)	Coefficients are functions of x

In this unit, we study **linear ODEs with constant coefficients** only – the most structured and solvable class.

3.1 Homogeneous Linear ODE

Definition: A linear ODE is called **homogeneous** if the right-hand side is **zero** (i.e., $f(x) = 0$).

$$f(D)y = 0 \quad \text{i.e.,} \quad a_0 \cdot D^n y + a_1 \cdot D^{(n-1)}y + \dots + a_n \cdot y = 0$$

Examples of homogeneous equations:

- $d^2y/dx^2 - 5 \cdot dy/dx + 6y = 0$
- $d^3y/dx^3 + 4 \cdot d^2y/dx^2 - 3y = 0$
- $(D^2 + 4D + 4)y = 0$

3.2 Non-Homogeneous Linear ODE

Definition: A linear ODE is called **non-homogeneous** if the right-hand side is **not zero** (i.e., $f(x) \neq 0$). The function $f(x)$ is called the **forcing function** or **input function**.

$$f(D)y = f(x) \quad \text{where } f(x) \neq 0$$

Examples of non-homogeneous equations:

- $d^2y/dx^2 - 3 \cdot dy/dx + 2y = e^{(2x)}$
- $d^2y/dx^2 + y = \sin x$
- $(D^2 - 4)y = x^2$

3.3 Why This Classification Matters

The general solution of a non-homogeneous ODE is built in two parts:

Structure of the General Solution:

General Solution (GS) = Complementary Function (CF) + Particular Integral (PI)

$$y = CF + PI$$

- **CF:** The complete solution of the associated homogeneous equation ($f(D)y = 0$). It contains n arbitrary constants for an n -th order ODE.
- **PI:** Any particular solution of the non-homogeneous equation ($f(D)y = f(x)$). It contains no arbitrary constants.

4.1 The Core Idea Behind the Auxiliary Equation

When we try to solve a homogeneous ODE with constant coefficients, we guess that the solution has the form:

$y = e^{mx}$

Why this guess? Because the derivative of e^{mx} is $m \cdot e^{mx}$, the second derivative is $m^2 \cdot e^{mx}$, and so on. This means if we substitute $y = e^{mx}$ into the ODE, every term will contain e^{mx} as a common factor, which we can cancel out.

4.2 Step-by-Step Formation of the Auxiliary Equation

Let us take the general second-order ODE:

$a \cdot d^2y/dx^2 + b \cdot dy/dx + c \cdot y = 0$

Step 1: Assume the trial solution $y = e^{mx}$
Then: $dy/dx = m \cdot e^{mx}$ and $d^2y/dx^2 = m^2 \cdot e^{mx}$

Step 2: Substitute into the ODE
 $a \cdot (m^2 \cdot e^{mx}) + b \cdot (m \cdot e^{mx}) + c \cdot (e^{mx}) = 0$

Step 3: Factor out e^{mx}
 $e^{mx} \cdot (a \cdot m^2 + b \cdot m + c) = 0$

Step 4: Since $e^{mx} \neq 0$ for any finite x , we must have:
 $a \cdot m^2 + b \cdot m + c = 0$ ← This is the Auxiliary Equation (AE)

Key Rule (MEMORIZE THIS):
To form the Auxiliary Equation, simply replace:
 $d^n y/dx^n$ by m^n (or equivalently, replace D by m in $f(D)$)
 $d^2y/dx^2 \rightarrow m^2$ $dy/dx \rightarrow m$ $y \rightarrow 1$ (m^0)
Then set the resulting polynomial = 0.

4.3 Examples of Forming the Auxiliary Equation

Given ODE	Auxiliary Equation (AE)
$d^2y/dx^2 - 5 \cdot dy/dx + 6y = 0$	$m^2 - 5m + 6 = 0$
$d^3y/dx^3 + 6 \cdot d^2y/dx^2 + 11 \cdot dy/dx + 6y = 0$	$m^3 + 6m^2 + 11m + 6 = 0$
$(D^2 + 4)y = 0$	$m^2 + 4 = 0$

$(D^4 - 1)y = 0$	$m^4 - 1 = 0$
$d^2y/dx^2 - 4 \cdot dy/dx + 4y = 0$	$m^2 - 4m + 4 = 0$

5.1 What is the Complementary Function (CF)?

Definition: The **Complementary Function** (CF) is the general solution of the associated **homogeneous** equation $f(D)y = 0$. It is formed by solving the Auxiliary Equation (AE) and writing the solution based on the nature of the roots.

The AE for an n-th order ODE is a polynomial of degree n in m, so it has exactly n roots (counting multiplicity, over complex numbers). The **nature of these roots** determines the form of the CF.

5.2 Complete Table of CF Cases

Case	Nature of Roots of AE	Roots	Form of CF
Case 1	Real and distinct (all different)	m_1, m_2 (distinct real)	$CF = C_1 \cdot e^{(m_1x)} + C_2 \cdot e^{(m_2x)}$
Case 2	Real, equal (repeated roots)	$m_1 = m_2 = m$ (double root)	$CF = (C_1 + C_2 \cdot x) \cdot e^{(mx)}$
Case 2b	Triple equal roots	$m_1 = m_2 = m_3 = m$	$CF = (C_1 + C_2 \cdot x + C_3 \cdot x^2) \cdot e^{(mx)}$
Case 3	Complex conjugate roots	$m = \alpha \pm i\beta$	$CF = e^{(\alpha x)} \cdot [C_1 \cdot \cos(\beta x) + C_2 \cdot \sin(\beta x)]$
Case 4	Complex repeated roots	$m = \alpha \pm i\beta$ (twice)	$CF = e^{(\alpha x)} \cdot [(C_1 + C_2 \cdot x) \cdot \cos(\beta x) + (C_3 + C_4 \cdot x) \cdot \sin(\beta x)]$
Case 5	Pure imaginary roots	$m = \pm i\beta$ ($\alpha = 0$)	$CF = C_1 \cdot \cos(\beta x) + C_2 \cdot \sin(\beta x)$
Mixed	Mix of distinct + repeated + complex	Various	Sum of corresponding terms for each root/pair

5.3 Detailed Explanation of Each Case

Case 1: Real and Distinct Roots

If the AE $m^2 + bm + c = 0$ has two different real roots m_1 and m_2 ($m_1 \neq m_2$), then the two independent solutions are $e^{(m_1x)}$ and $e^{(m_2x)}$. The general CF is their linear combination:

$CF = C_1 \cdot e^{(m_1x)} + C_2 \cdot e^{(m_2x)}$

For n distinct real roots $m_1, m_2, \dots, m_n$:

$CF = C_1 \cdot e^{(m_1x)} + C_2 \cdot e^{(m_2x)} + \dots + C_n \cdot e^{(m_nx)}$

Case 2: Repeated (Equal) Roots

If the AE has a repeated root $m_1 = m_2 = m$ (double root), we cannot simply write $C_1 \cdot e^{(mx)} + C_2 \cdot e^{(mx)}$ since these are not independent (they combine into one arbitrary constant). Instead, the theory shows the independent solutions are $e^{(mx)}$ and $x \cdot e^{(mx)}$:

$$CF = (C_1 + C_2 \cdot x) \cdot e^{(mx)}$$

For a triple root m (root of multiplicity 3): $CF = (C_1 + C_2 \cdot x + C_3 \cdot x^2) \cdot e^{(mx)}$

For a root of multiplicity k : $CF = (C_1 + C_2 \cdot x + C_3 \cdot x^2 + \dots + C_k \cdot x^{(k-1)}) \cdot e^{(mx)}$

Case 3: Complex Conjugate Roots

If the AE has complex roots $m = \alpha + i\beta$ and $m = \alpha - i\beta$ (complex conjugates always appear in pairs for real coefficient equations), then using Euler's formula $e^{(i\theta)} = \cos \theta + i \cdot \sin \theta$, the two independent real solutions are $e^{(\alpha x)} \cdot \cos(\beta x)$ and $e^{(\alpha x)} \cdot \sin(\beta x)$:

$$CF = e^{(\alpha x)} \cdot [C_1 \cdot \cos(\beta x) + C_2 \cdot \sin(\beta x)]$$

Special Sub-case: Pure Imaginary Roots ($\alpha = 0$)

If $m = \pm i\beta$ (i.e., $\alpha = 0$), then:

$$CF = C_1 \cdot \cos(\beta x) + C_2 \cdot \sin(\beta x) \quad (\text{no } e^{(\alpha x)} \text{ factor since } e^0 = 1)$$

Section 6 Particular Integral – Concept and Justification The Second Piece

6.1 What is the Particular Integral (PI)?

Definition: A **Particular Integral** (PI) is any one specific solution of the non-homogeneous ODE $f(D)y = f(x)$ that does NOT contain any arbitrary constants.

The PI "accounts for" the right-hand side forcing function $f(x)$. It is found by operating on $f(x)$ with the inverse operator $1/f(D)$:

$$PI = [1/f(D)] \cdot f(x)$$

6.2 Why Do We Need Both CF and PI?

Think of it this way:

- The CF solves the homogeneous part (the "natural" behavior of the system with no external input)
- The PI solves the non-homogeneous part (the "forced" response due to the external input $f(x)$)
- Adding them together gives ALL solutions to the ODE

Superposition Theorem: If $y_c = CF$ is the general solution of $f(D)y = 0$, and $y_p = PI$ is any particular solution of $f(D)y = f(x)$, then the **general solution** of $f(D)y = f(x)$ is:

$$y = y_c + y_p = CF + PI$$

7.1 The Inverse Operator Method

The symbol $1/f(D)$ means "apply the inverse of the operator $f(D)$ ". Different types of $f(x)$ require different operator manipulation techniques. Below are the complete standard results.

7.2 Type 1: $f(x) = e^{ax}$

Standard Result: If $f(a) \neq 0$, then:

$$PI = [1/f(D)] \cdot e^{ax} = e^{ax} / f(a)$$

Rule: Simply replace D by a everywhere in $f(D)$, provided $f(a) \neq 0$.

If $f(a) = 0$ (failure case): Then a is a root of $f(D)=0$, and we use:

$$PI = x \cdot e^{ax} / f(a) \quad (\text{if } f(a) \neq 0)$$

$$PI = x^2 \cdot e^{ax} / f'(a) \quad (\text{if } f(a) = 0 \text{ too), and so on.}$$

7.3 Type 2: $f(x) = \sin(ax)$ or $\cos(ax)$

Standard Result: Replace D^2 by $-a^2$ (since $D \cdot \sin(ax) = a \cdot \cos(ax)$, $D^2 \cdot \sin(ax) = -a^2 \cdot \sin(ax)$)

$$PI = [1/f(D^2)] \cdot \sin(ax) \rightarrow \text{replace } D^2 \text{ by } -a^2 \text{ in } f(D^2), \text{ provided } f(-a^2) \neq 0$$

Similarly for $\cos(ax)$.

If $f(-a^2) = 0$ (failure):

$$PI = x \cdot [1/f(D^2)] \cdot \sin(ax) \text{ (multiply by } x \text{ and differentiate denominator)}$$

7.4 Type 3: $f(x) = x^n$ (polynomial)

Method: Express $1/f(D)$ as a power series in D (using binomial expansion), then apply term-by-term to x^n . Since $D^k(x^n) = 0$ when $k > n$, the series terminates.

$$\text{Example: } PI = 1/(1+D^2) \cdot x^3 = (1 - D^2 + D^4 - \dots) \cdot x^3 = x^3 - 6x \text{ (since } D^2x^3=6x, D^4x^3=0)$$

7.5 Type 4: $f(x) = e^{ax} \cdot V(x)$ (exponential shift)

Exponential Shift Rule:

$$\left[\frac{1}{f(D)} \right] \cdot e^{ax} \cdot V(x) = e^{ax} \cdot \left[\frac{1}{f(D+a)} \right] \cdot V(x)$$

This allows us to "shift" the exponential out, solve for $V(x)$ by the appropriate method, then multiply back by e^{ax} .

7.6 Type 5: $f(x) = x \cdot \sin(ax)$ or $x \cdot \cos(ax)$

Method: Use the formula:

$$\left[\frac{1}{f(D)} \right] \cdot x \cdot \sin(ax) = \text{Imaginary part of } \left[\frac{1}{f(D)} \right] \cdot x \cdot e^{iax}$$

Then apply the exponential shift to handle $x \cdot e^{iax}$, expand using binomial, separate real and imaginary parts.

7.7 Type 6: General $f(x)$ – Method of Partial Fractions

Method: Decompose $1/f(D)$ into partial fractions in terms of $(D - m_i)$, then apply each term to $f(x)$ separately. This works when $f(x)$ is any function that can be integrated.

7.8 Complete PI Reference Table

Type of $f(x)$	PI Formula	Condition / Failure Rule
e^{ax}	$PI = e^{ax} / f(a)$	If $f(a)=0$: $PI = x \cdot e^{ax} / f'(a)$; if $f'(a)=0$: $PI = x^2 \cdot e^{ax} / f''(a)$
$\sin(ax)$	$PI = \sin(ax) / f(-a^2)$	Replace $D^2 \rightarrow -a^2$; if $f(-a^2)=0$, multiply by x and differentiate
$\cos(ax)$	$PI = \cos(ax) / f(-a^2)$	Same as sin rule; replace $D^2 \rightarrow -a^2$
x^n (polynomial)	Expand $1/f(D)$ in powers of D via binomial	Terminate when $D^{(n+1)}$ acts on $x^n \rightarrow 0$
$e^{ax} \cdot V(x)$	$PI = e^{ax} \cdot \left[\frac{1}{f(D+a)} \right] \cdot V(x)$	Exponential shift theorem
$x \cdot \sin(ax)$	Use $\text{Im}\{ \left[\frac{1}{f(D)} \right] \cdot x \cdot e^{iax} \}$	Exponential shift then binomial
General $f(x)$	Partial fractions or variation of parameters	Always works

Section 8 General Solution = CF + PI Putting It All Together

8.1 The Complete Solution Structure

Complete General Solution:

For $f(D)y = f(x)$:

$$y = CF + PI$$

where CF contains n arbitrary constants (for an n-th order equation) and PI has no arbitrary constants.

The n arbitrary constants in CF are determined if n **initial conditions** or **boundary conditions** are given.

8.2 Step-by-Step Process for ANY Problem

MASTER ALGORITHM – Apply to every problem

1. **Write the ODE** in standard form using D-notation
2. **Form the AE** by replacing D with m, set = 0
3. **Solve the AE** – find all n roots
4. **Write CF** using the root-case table (Section 5.2)
5. **Find PI** using the appropriate method from Section 7
6. **Write General Solution:** $y = CF + PI$
7. **Apply ICs/BCs** if given to find constant values

9.1 Definition of the Wronskian

Definition: Given n functions $y_1, y_2, \dots, y_n$ that are $(n-1)$ times differentiable, the **Wronskian** $W(y_1, y_2, \dots, y_n)$ is the determinant:

$$W(y_1, y_2, \dots, y_n) = \begin{vmatrix} y_1 & y_2 & \dots & y_n \\ y_1' & y_2' & \dots & y_n' \\ y_1'' & y_2'' & \dots & y_n'' \\ \vdots & \vdots & \ddots & \vdots \\ y_1^{(n-1)} & y_2^{(n-1)} & \dots & y_n^{(n-1)} \end{vmatrix}$$

9.2 Wronskian and Linear Independence

Theorem (Abel's Theorem):

If $y_1, y_2, \dots, y_n$ are solutions of the linear ODE $f(D)y = 0$, then:

- $W \neq 0$ for some $x \rightarrow$ the solutions are **linearly independent** (form a fundamental set)
- $W = 0$ for all $x \rightarrow$ the solutions are **linearly dependent**

9.3 Calculating Wronskian – Step by Step

Example: Check if $y_1 = e^x$ and $y_2 = e^{(2x)}$ are linearly independent

Step 1: Write the Wronskian determinant (2×2 for 2 functions)

$$W(y_1, y_2) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} e^x & e^{(2x)} \\ e^x & 2e^{(2x)} \end{vmatrix}$$

Step 2: Compute the 2×2 determinant

$$W = y_1 \cdot y_2' - y_2 \cdot y_1' = e^x \cdot 2e^{(2x)} - e^{(2x)} \cdot e^x = 2e^{(3x)} - e^{(3x)} = e^{(3x)}$$

Step 3: Interpret the result

$$W = e^{(3x)} \neq 0 \text{ for any real } x$$

$\therefore y_1 = e^x$ and $y_2 = e^{(2x)}$ are LINEARLY INDEPENDENT

10.1 Why Variation of Parameters?

The operator method works well for standard $f(x)$ like exponentials, trigonometric, and polynomials. But for more general $f(x)$ (like $\tan x$, $\sec x$, $1/x$, etc.), we need a more powerful method called **Variation of Parameters**.

10.2 Method for Second-Order ODE

Consider the ODE: $d^2y/dx^2 + P \cdot dy/dx + Q \cdot y = R(x)$, where P and Q are constants (or functions of x).

Let $CF = C_1 \cdot y_1 + C_2 \cdot y_2$, where y_1 and y_2 are two linearly independent solutions.

Variation of Parameters Formulas:

$$PI = -y_1 \cdot \int (y_2 \cdot R / W) \, dx + y_2 \cdot \int (y_1 \cdot R / W) \, dx$$

where $W =$ Wronskian of y_1 and $y_2 = y_1 \cdot y_2' - y_2 \cdot y_1'$

Derivation: We assume $PI = A(x) \cdot y_1 + B(x) \cdot y_2$ (vary the constants C_1, C_2 as functions of x), then impose conditions to determine $A'(x)$ and $B'(x)$, giving the formulas above.

10.3 Step-by-Step Procedure

Step 1

Find $CF = C_1 \cdot y_1 + C_2 \cdot y_2$ by solving the homogeneous equation.

Step 2

Compute Wronskian: $W = y_1 \cdot y_2' - y_2 \cdot y_1'$

Step 3

Compute $A'(x) = -y_2 \cdot R(x)/W$ and $B'(x) = y_1 \cdot R(x)/W$

Step 4

Integrate: $A(x) = \int A'(x) \, dx$ and $B(x) = \int B'(x) \, dx$

Step 5

$$PI = A(x) \cdot y_1 + B(x) \cdot y_2$$

Step 6

General Solution: $y = CF + PI$

Section 11 Simultaneous Linear Differential Equations Two or More Unknowns

11.1 What are Simultaneous Linear ODEs?

Definition: A system of linear ODEs involves two or more unknown functions of the same independent variable x (or t), and each ODE contains derivatives of these unknowns. The unknowns are found simultaneously.

The standard form for two unknowns $x(t)$ and $y(t)$:

$$f_1(D)x + g_1(D)y = h_1(t) \quad f_2(D)x + g_2(D)y = h_2(t) \quad \text{where } D = d/dt$$

11.2 Method of Elimination

Step 1

Treat the system like simultaneous algebraic equations, with D as the "variable". Use elimination to get a single ODE in one unknown (say x).

Step 2

Solve the resulting ODE for x using the standard CF+PI method.

Step 3

Substitute x back into one of the original equations to find y .

Step 4

Note: The arbitrary constants in the solutions for x and y are NOT all independent – apply the system equations to find relationships between them.

Section 12 Applications – LCR Circuits Electrical Engineering Application

12.1 The LCR Circuit

A series LCR circuit consists of:

- **L** = Inductance (in Henries, H) – opposes change in current
- **C** = Capacitance (in Farads, F) – stores charge
- **R** = Resistance (in Ohms, Ω) – dissipates energy
- **E(t)** = Applied EMF (Electromotive Force)

12.2 Governing ODE for LCR Circuit

Applying Kirchhoff's Voltage Law (sum of voltage drops = applied EMF) and using the relations:

- Voltage across R = $R \cdot i$ (Ohm's law)
- Voltage across L = $L \cdot di/dt$ (inductor equation)
- Voltage across C = $q/C = (1/C) \cdot \int i \, dt$ (capacitor equation)

$L \cdot d^2q/dt^2 + R \cdot dq/dt + (1/C) \cdot q = E(t)$ where q = charge, $i = dq/dt$ = current OR equivalently in terms of current i : $L \cdot di/dt + R \cdot i + (1/C) \cdot \int i \, dt = E(t)$
Differentiating with respect to t : $L \cdot d^2i/dt^2 + R \cdot di/dt + (1/C) \cdot i = dE/dt$

12.3 Analogies with Mechanical Systems

Electrical Quantity	Symbol	Mechanical Analogue
Charge	q	Displacement x
Current	$i = dq/dt$	Velocity dx/dt
Inductance	L	Mass m
Resistance	R	Damping coefficient c
1/Capacitance	$1/C$	Spring constant k
EMF	$E(t)$	Applied force $F(t)$

Section 13 Applications – Simple Harmonic Motion Oscillation and Vibration

13.1 What is Simple Harmonic Motion (SHM)?

Definition: Simple Harmonic Motion is the motion of a particle such that the restoring force is directly proportional to the displacement from the mean position and directed toward it.

$F = -kx$ (restoring force $\propto$ displacement, directed inward) By Newton's second law: $m \cdot d^2x/dt^2 = -kx \rightarrow d^2x/dt^2 + (k/m) \cdot x = 0$ Let $\omega^2 = k/m$ (angular frequency) $\rightarrow d^2x/dt^2 + \omega^2 \cdot x = 0$ ← Standard SHM equation

13.2 Solution of SHM Equation

The AE for $d^2x/dt^2 + \omega^2 \cdot x = 0$ is:

$$m^2 + \omega^2 = 0 \rightarrow m = \pm i\omega \text{ (pure imaginary roots)}$$

$\therefore$ CF (General Solution) = $C_1 \cdot \cos(\omega t) + C_2 \cdot \sin(\omega t)$

This can be written in amplitude-phase form as:

$$x(t) = A \cdot \cos(\omega t + \phi) \qquad \text{where } A = \text{amplitude, } \phi = \text{initial phase}$$

13.3 Damped and Forced Oscillations

Type	ODE	Nature
Undamped Free	$d^2x/dt^2 + \omega^2x = 0$	Oscillates forever (no energy loss)
Damped Free	$d^2x/dt^2 + 2k \cdot dx/dt + \omega^2x = 0$	Decays over time
Undamped Forced	$d^2x/dt^2 + \omega^2x = F \cdot \cos(pt)$	Resonance if $p = \omega$
Damped Forced	$d^2x/dt^2 + 2k \cdot dx/dt + \omega^2x = F \cdot \cos(pt)$	Steady-state oscillation

● Example 1: CF with Real Distinct Roots**Problem:** Solve $d^2y/dx^2 - 5 \cdot dy/dx + 6y = 0$ **Step 1: Identify the type of equation**The right-hand side = 0, so this is a **homogeneous linear ODE with constant coefficients** of order 2.

General solution = CF only (no PI needed).

Step 2: Write in D-notation

$$(D^2 - 5D + 6)y = 0 \quad \text{where } D = d/dx$$

Step 3: Form the Auxiliary Equation

Replace D by m: $m^2 - 5m + 6 = 0$

Explanation: We assume $y = e^{(mx)}$; substituting gives $(m^2 - 5m + 6) \cdot e^{(mx)} = 0$; since $e^{(mx)} \neq 0$, we get $m^2 - 5m + 6 = 0$.**Step 4: Solve the Auxiliary Equation by factorization**

$$m^2 - 5m + 6 = 0$$

We need two numbers that multiply to +6 and add to -5. Those are -2 and -3.

$$\therefore m^2 - 2m - 3m + 6 = 0$$

$$\rightarrow m(m - 2) - 3(m - 2) = 0$$

$$\rightarrow (m - 2)(m - 3) = 0$$

$$\therefore m = 2 \quad \text{or} \quad m = 3 \quad \leftarrow \text{Two REAL, DISTINCT roots}$$

Step 5: Write the Complementary FunctionSince roots $m_1 = 2$ and $m_2 = 3$ are real and distinct (Case 1):

$$CF = C_1 \cdot e^{(m_1x)} + C_2 \cdot e^{(m_2x)} = C_1 \cdot e^{(2x)} + C_2 \cdot e^{(3x)}$$

Step 6: Write the General SolutionSince the equation is homogeneous, $PI = 0$.

$$\therefore y = CF + PI = C_1 \cdot e^{(2x)} + C_2 \cdot e^{(3x)} + 0$$

General Solution: $y = C_1 \cdot e^{(2x)} + C_2 \cdot e^{(3x)}$

● Example 2: CF with Repeated Roots

Problem: Solve $d^2y/dx^2 - 4 \cdot dy/dx + 4y = 0$

Step 1: Type

Homogeneous linear ODE, order 2. In D-notation: $(D^2 - 4D + 4)y = 0$.

Step 2: Form Auxiliary Equation

Replace D by m: $m^2 - 4m + 4 = 0$

Step 3: Solve AE

$$m^2 - 4m + 4 = 0$$

Recognize this as a perfect square: $(m - 2)^2 = 0$

$\therefore m = 2, 2 \leftarrow$ A REPEATED root (double root), $m = 2$

Step 4: Write CF for repeated root (Case 2)

Since $m_1 = m_2 = 2$ (double root):

$$CF = (C_1 + C_2 \cdot x) \cdot e^{(2x)}$$

Explanation: We cannot write $C_1 \cdot e^{(2x)} + C_2 \cdot e^{(2x)}$ because they combine into one arbitrary constant. The correct independent pair of solutions for a double root m is $e^{(mx)}$ and $x \cdot e^{(mx)}$.

$$\text{General Solution: } y = (C_1 + C_2 \cdot x) \cdot e^{(2x)}$$

● Example 3: CF with Complex Roots

Problem: Solve $d^2y/dx^2 + 4y = 0$

Step 1: Type and D-notation

Homogeneous, order 2. D-notation: $(D^2 + 4)y = 0$.

Step 2: Form and Solve AE

$$\text{AE: } m^2 + 4 = 0$$

$$\rightarrow m^2 = -4$$

$$\rightarrow m = \pm\sqrt{-4} = \pm\sqrt{4}\cdot\sqrt{-1} = \pm 2i$$

So $m = 0 + 2i$ and $m = 0 - 2i \rightarrow \alpha = 0, \beta = 2$ (pure imaginary roots, Case 5)

Step 3: Write CF for complex roots $\alpha \pm i\beta$ with $\alpha = 0, \beta = 2$

$$\text{CF} = e^{(\alpha x)} \cdot [C_1 \cdot \cos(\beta x) + C_2 \cdot \sin(\beta x)]$$

$$= e^{(0 \cdot x)} \cdot [C_1 \cdot \cos(2x) + C_2 \cdot \sin(2x)]$$

$$= 1 \cdot [C_1 \cdot \cos(2x) + C_2 \cdot \sin(2x)]$$

$$\text{General Solution: } y = C_1 \cdot \cos(2x) + C_2 \cdot \sin(2x)$$

● Example 4: Third-Order ODE – Three Distinct Real Roots

Problem: Solve $d^3y/dx^3 - 6 \cdot d^2y/dx^2 + 11 \cdot dy/dx - 6y = 0$

Step 1: D-notation

$$(D^3 - 6D^2 + 11D - 6)y = 0$$

Step 2: Form AE

$$m^3 - 6m^2 + 11m - 6 = 0$$

Step 3: Solve AE – try rational roots

Try $m = 1$: $1 - 6 + 11 - 6 = 0 \checkmark \rightarrow (m - 1)$ is a factor

Divide: $m^3 - 6m^2 + 11m - 6 = (m - 1)(m^2 - 5m + 6)$

Factor $m^2 - 5m + 6$: two numbers multiplying to 6, adding to -5 : are $-2, -3$

$$\rightarrow (m - 2)(m - 3)$$

$$\text{So: } (m - 1)(m - 2)(m - 3) = 0$$

$\therefore m = 1, 2, 3 \leftarrow$ Three distinct real roots

Step 4: CF (Case 1 extended)

$$CF = C_1 \cdot e^x + C_2 \cdot e^{2x} + C_3 \cdot e^{3x}$$

$$\text{General Solution: } y = C_1 \cdot e^x + C_2 \cdot e^{2x} + C_3 \cdot e^{3x}$$

● Example 5: Mixed Roots (Real + Complex)

Problem: Solve $(D^3 + D^2 + D + 1)y = 0$

Step 1: Form AE

$$m^3 + m^2 + m + 1 = 0$$

Step 2: Factor by grouping

$$m^2(m + 1) + 1(m + 1) = 0$$

$$(m + 1)(m^2 + 1) = 0$$

$$\therefore m = -1 \quad \text{or} \quad m^2 = -1 \rightarrow m = \pm i$$

Roots: $m = -1$ (real, distinct), $m = 0 \pm i$ (complex, $\alpha=0$, $\beta=1$)

Step 3: CF

From $m = -1$: $C_1 \cdot e^{-x}$

From $m = \pm i$: $C_2 \cdot \cos(x) + C_3 \cdot \sin(x)$

$$\text{CF} = C_1 \cdot e^{-x} + C_2 \cdot \cos(x) + C_3 \cdot \sin(x)$$

$$\text{General Solution: } y = C_1 \cdot e^{-x} + C_2 \cdot \cos(x) + C_3 \cdot \sin(x)$$

● Example 6: PI for $f(x) = e^{(ax)}$ – Standard Case

Problem: Solve $(D^2 - 3D + 2)y = e^{(3x)}$

Step 1: Type

Non-homogeneous. $f(x) = e^{(3x)}$. Need CF and PI.

Step 2: Find CF

AE: $m^2 - 3m + 2 = 0 \rightarrow (m - 1)(m - 2) = 0 \rightarrow m = 1, 2$

CF = $C_1 \cdot e^x + C_2 \cdot e^{(2x)}$

Step 3: Find PI

PI = $[1/(D^2 - 3D + 2)] \cdot e^{(3x)}$

Rule: For $e^{(ax)}$, replace D by a = 3 in the operator polynomial:

PI = $e^{(3x)} / (3^2 - 3 \cdot 3 + 2) = e^{(3x)} / (9 - 9 + 2) = e^{(3x)} / 2$

Step 4: Verify $f(a) \neq 0$

$f(3) = 9 - 9 + 2 = 2 \neq 0$. ✓ Standard case applies. No failure.

Step 5: General Solution

$y = CF + PI = C_1 \cdot e^x + C_2 \cdot e^{(2x)} + (1/2) \cdot e^{(3x)}$

$$y = C_1 \cdot e^x + C_2 \cdot e^{(2x)} + (1/2) \cdot e^{(3x)}$$

● Example 7: PI Failure Case for $e^{(ax)}$

Problem: Solve $(D^2 - 3D + 2)y = e^{(2x)}$

Step 1: CF (same as Example 6)

$$\text{AE: } m^2 - 3m + 2 = 0 \rightarrow m = 1, 2. \quad \text{CF} = C_1 \cdot e^x + C_2 \cdot e^{(2x)}$$

Step 2: Attempt PI using standard rule

$$\text{PI} = e^{(2x)} / f(2) = e^{(2x)} / (4 - 6 + 2) = e^{(2x)} / 0 \leftarrow \text{FAILURE! Division by zero!}$$

Step 3: Apply Failure Rule (multiply by x, differentiate denominator)

Since $f(2) = 0$ (i.e., $m = 2$ is a root of AE), use:

$$\text{PI} = x \cdot e^{(ax)} / f'(a) \text{ where } f'(D) = 2D - 3$$

$$f'(2) = 2(2) - 3 = 4 - 3 = 1$$

$$\therefore \text{PI} = x \cdot e^{(2x)} / 1 = x \cdot e^{(2x)}$$

Step 4: Verify

Check that $f'(a) \neq 0$: $f'(2) = 1 \neq 0$. ✓ No further failure.

Step 5: General Solution

$$y = \text{CF} + \text{PI} = C_1 \cdot e^x + C_2 \cdot e^{(2x)} + x \cdot e^{(2x)}$$

$$y = C_1 \cdot e^x + C_2 \cdot e^{(2x)} + x \cdot e^{(2x)}$$

● Example 8: PI for $\sin(ax)$ – Standard Case

Problem: Solve $(D^2 + 5D + 6)y = \sin(2x)$

Step 1: CF

$$\text{AE: } m^2 + 5m + 6 = 0 \rightarrow (m + 2)(m + 3) = 0 \rightarrow m = -2, -3$$

$$\text{CF} = C_1 \cdot e^{(-2x)} + C_2 \cdot e^{(-3x)}$$

Step 2: PI for $\sin(2x)$

$$\text{PI} = [1/(D^2 + 5D + 6)] \cdot \sin(2x)$$

Key rule: For $\sin(ax)$, replace D^2 by $-a^2 = -(2^2) = -4$

BUT we cannot replace D alone (only D^2 can be replaced directly in $\sin/\cos$ type).

First check if the operator is a function of D^2 only. Here we have $5D$ term, so we proceed differently.

Step 3: Handle the D term in denominator

$$\text{PI} = [1/(D^2 + 5D + 6)] \cdot \sin(2x)$$

Replace $D^2 \rightarrow -4$:

$$= [1/(-4 + 5D + 6)] \cdot \sin(2x)$$

$$= [1/(5D + 2)] \cdot \sin(2x)$$

Now rationalize: multiply numerator and denominator by $(2 - 5D)$:

$$= [(2 - 5D) / ((2 + 5D)(2 - 5D))] \cdot \sin(2x)$$

$$= [(2 - 5D) / (4 - 25D^2)] \cdot \sin(2x)$$

Replace $D^2 \rightarrow -4$ again in denominator:

$$= [(2 - 5D) / (4 - 25(-4))] \cdot \sin(2x)$$

$$= [(2 - 5D) / (4 + 100)] \cdot \sin(2x)$$

$$= [(2 - 5D) / 104] \cdot \sin(2x)$$

Step 4: Apply operator to $\sin(2x)$

$$(2 - 5D) \cdot \sin(2x) = 2 \cdot \sin(2x) - 5 \cdot D \cdot \sin(2x)$$

$$D \cdot \sin(2x) = 2 \cdot \cos(2x) \text{ (differentiating } \sin(2x))$$

$$= 2 \cdot \sin(2x) - 5 \cdot (2 \cdot \cos(2x))$$

$$= 2 \cdot \sin(2x) - 10 \cdot \cos(2x)$$

Step 5: Write PI

$$\text{PI} = [2 \cdot \sin(2x) - 10 \cdot \cos(2x)] / 104 = (1/52) \cdot \sin(2x) - (5/52) \cdot \cos(2x)$$

$$y = C_1 \cdot e^{(-2x)} + C_2 \cdot e^{(-3x)} + (1/52) \cdot \sin(2x) - (5/52) \cdot \cos(2x)$$

● Example 9: PI for Polynomial $f(x) = x^2$

Problem: Solve $(D^2 + D)y = x^2$

Step 1: CF

$$\text{AE: } m^2 + m = 0 \rightarrow m(m + 1) = 0 \rightarrow m = 0, -1$$

$$\text{CF} = C_1 \cdot e^{(0 \cdot x)} + C_2 \cdot e^{(-x)} = C_1 + C_2 \cdot e^{(-x)} \quad (\text{since } e^0 = 1)$$

Step 2: PI for x^2

$$\text{PI} = [1/(D^2 + D)] \cdot x^2$$

$$\text{Factor out D: PI} = [1/D] \cdot [1/(D + 1)] \cdot x^2$$

Step 3: Expand $[1/(1+D)]$ using binomial series

$$[1/(1+D)] = (1 + D)^{-1} = 1 - D + D^2 - D^3 + \dots$$

(This series terminates when applied to x^2 since $D^3 x^2 = 0$)

Step 4: Apply to x^2

$$[1/(1+D)] \cdot x^2 = (1 - D + D^2 - \dots) \cdot x^2$$

$$= x^2 - D(x^2) + D^2(x^2) - \dots$$

$$D(x^2) = 2x$$

$$D^2(x^2) = D(2x) = 2$$

$$D^3(x^2) = 0$$

$$= x^2 - 2x + 2$$

Step 5: Apply $[1/D] = \int \dots dx$ (integration)

$$\text{PI} = [1/D] \cdot (x^2 - 2x + 2) = \int (x^2 - 2x + 2) dx$$

$$= x^3/3 - x^2 + 2x$$

$$y = C_1 + C_2 \cdot e^{(-x)} + x^3/3 - x^2 + 2x$$

● Example 10: PI with Exponential Shift ($e^x \cdot \sin x$)

Problem: Solve $(D^2 - 2D + 5)y = e^x \cdot \sin(2x)$

Step 1: CF

$$\text{AE: } m^2 - 2m + 5 = 0$$

$$\text{Discriminant: } b^2 - 4ac = 4 - 20 = -16 < 0$$

$$m = [2 \pm \sqrt{(-16)}] / 2 = [2 \pm 4i] / 2 = 1 \pm 2i$$

$\alpha = 1, \beta = 2 \rightarrow$ Complex roots

$$\text{CF} = e^x \cdot [C_1 \cdot \cos(2x) + C_2 \cdot \sin(2x)]$$

Step 2: PI using Exponential Shift Rule

$$\text{PI} = [1/(D^2 - 2D + 5)] \cdot e^x \cdot \sin(2x)$$

Exponential Shift: $[1/f(D)] \cdot e^{ax} \cdot V(x) = e^{ax} \cdot [1/f(D+a)] \cdot V(x)$ with $a = 1$

$$\text{PI} = e^x \cdot [1/((D+1)^2 - 2(D+1) + 5)] \cdot \sin(2x)$$

Step 3: Expand the shifted operator

$$(D+1)^2 - 2(D+1) + 5$$

$$= D^2 + 2D + 1 - 2D - 2 + 5$$

$$= D^2 + 4$$

$$\therefore \text{PI} = e^x \cdot [1/(D^2 + 4)] \cdot \sin(2x)$$

Step 4: Apply to $\sin(2x)$: replace $D^2 \rightarrow -(2^2) = -4$

$$[1/(D^2 + 4)] \cdot \sin(2x) = \sin(2x) / (-4 + 4) = \sin(2x) / 0 \leftarrow \text{FAILURE!}$$

$m = 2i$ satisfies $D^2 + 4 = 0$, so this is a failure case.

Step 5: Failure rule for sin/cos – multiply by x and differentiate

$$\text{PI (inner)} = [1/(D^2 + 4)] \cdot \sin(2x)$$

$$= x \cdot [1/(2D)] \cdot \sin(2x) \quad (\text{differentiating denominator } D^2 + 4 \text{ w.r.t. } D^2 \text{ gives } 1, \text{ multiply by } x/(2D \text{ with chain}))$$

$$\text{Actually use: } \text{PI} = \text{Im} \{ [1/(D^2 + 4)] \cdot e^{(2ix)} \} = \text{Im} \{ x \cdot e^{(2ix)} / (2 \cdot 2i) \} = \text{Im} \{ x \cdot e^{(2ix)} / (4i) \}$$

$$e^{(2ix)} = \cos(2x) + i \cdot \sin(2x)$$

$$x \cdot e^{(2ix)} / (4i) = x(\cos(2x) + i \cdot \sin(2x)) / (4i) = x(\cos(2x) + i \cdot \sin(2x)) \cdot (-i) / 4$$

$$\text{Im part} = x \cdot [-\cos(2x)] / 4 = -x \cdot \cos(2x) / 4$$

$$\therefore \text{Inner PI} = -x \cdot \cos(2x) / 4$$

Step 6: Complete PI

$$\text{PI} = e^x \cdot (-x \cdot \cos(2x) / 4) = -(x/4) \cdot e^x \cdot \cos(2x)$$

$$y = e^x \cdot [C_1 \cdot \cos(2x) + C_2 \cdot \sin(2x)] - (x/4) \cdot e^x \cdot \cos(2x)$$

● Example 11: Fourth Order – Two Pairs of Repeated Complex Roots

Problem: Solve $(D^2 + 4)^2 y = 0$

AE

$$(m^2 + 4)^2 = 0 \rightarrow m^2 + 4 = 0 \text{ (twice)} \rightarrow m = \pm 2i \text{ (each repeated twice)}$$

Roots

$$m = +2i \text{ (double)}, m = -2i \text{ (double)}; \alpha = 0, \beta = 2$$

CF (Case 4 with $\alpha=0$)

$$CF = (C_1 + C_2 \cdot x) \cdot \cos(2x) + (C_3 + C_4 \cdot x) \cdot \sin(2x)$$

$$y = (C_1 + C_2 x) \cdot \cos(2x) + (C_3 + C_4 x) \cdot \sin(2x)$$

● Example 12: PI for $\cos(ax)$ using D^2 Replacement

Problem: Solve $(D^2 + 9)y = \cos(3x)$

CF

$$AE: m^2 + 9 = 0 \rightarrow m = \pm 3i. CF = C_1 \cdot \cos(3x) + C_2 \cdot \sin(3x)$$

PI attempt

$$\text{Replace } D^2 \rightarrow -9: PI = \cos(3x)/(-9+9) = \cos(3x)/0 \leftarrow \text{FAILURE (resonance case)}$$

Failure Rule for $\cos(ax)$

$$PI = \text{Re}\{[1/(D^2+9)] \cdot e^{(3ix)}\} \text{ with failure.}$$

$$= \text{Re}\{x \cdot e^{(3ix)} / (d/d(D^2)(D^2+9)) \cdot 2 \cdot D\} \text{ [using } x/(2iD) \text{ shortcut]}$$

$$= \text{Re}\{x \cdot e^{(3ix)} / (2 \cdot 3i)\} = \text{Re}\{x(\cos 3x + i \sin 3x) / (6i)\}$$

$$= \text{Re}\{x(\cos 3x + i \sin 3x) \cdot (-i) / 6\}$$

$$= \text{Re}\{x[-i \cos 3x + \sin 3x] / 6\}$$

$$= x \cdot \sin(3x) / 6$$

$$y = C_1 \cdot \cos(3x) + C_2 \cdot \sin(3x) + (x/6) \cdot \sin(3x)$$

● Example 13: PI for $f(x) = x^3$ (Polynomial)

Problem: Solve $(D^2 + 1)y = x^3$

CF

$$m^2 + 1 = 0 \rightarrow m = \pm i. \text{ CF} = C_1 \cdot \cos(x) + C_2 \cdot \sin(x)$$

PI

$$\text{PI} = [1/(1+D^2)] \cdot x^3$$

$$= (1+D^2)^{-1} \cdot x^3$$

$$= (1 - D^2 + D^4 - \dots) \cdot x^3$$

$$D^2(x^3) = D(3x^2) = 6x$$

$$D^4(x^3) = D^2(6x) = D(6) = 0$$

$$\therefore \text{PI} = x^3 - 6x$$

$$y = C_1 \cdot \cos(x) + C_2 \cdot \sin(x) + x^3 - 6x$$

● Example 14: Initial Value Problem

Problem: Solve $d^2y/dx^2 + y = 0$, $y(0) = 2$, $y'(0) = -3$

CF (no PI needed, RHS = 0)

$$\text{AE: } m^2+1=0 \rightarrow m=\pm i. \text{ CF} = C_1 \cdot \cos(x) + C_2 \cdot \sin(x)$$

Apply $y(0) = 2$

$$y(0) = C_1 \cdot \cos(0) + C_2 \cdot \sin(0) = C_1 \cdot 1 + C_2 \cdot 0 = C_1 = 2. \text{ So } C_1 = 2.$$

Differentiate y

$$y' = -C_1 \cdot \sin(x) + C_2 \cdot \cos(x)$$

Apply $y'(0) = -3$

$$y'(0) = -C_1 \cdot 0 + C_2 \cdot 1 = C_2 = -3. \text{ So } C_2 = -3.$$

$$y = 2 \cdot \cos(x) - 3 \cdot \sin(x)$$

Example 15: Variation of Parameters

Problem: Solve $d^2y/dx^2 + y = \sec(x)$ using Variation of Parameters

Step 1: CF

$$\text{AE: } m^2 + 1 = 0 \rightarrow m = \pm i. \quad y_1 = \cos(x), \quad y_2 = \sin(x). \quad \text{CF} = C_1 \cdot \cos(x) + C_2 \cdot \sin(x)$$

Step 2: Wronskian

$$\begin{aligned} W &= y_1 \cdot y_2' - y_2 \cdot y_1' = \cos(x) \cdot \cos(x) - \sin(x) \cdot (-\sin(x)) \\ &= \cos^2(x) + \sin^2(x) = 1 \end{aligned}$$

Step 3: $A'(x)$ and $B'(x)$

$$R(x) = \sec(x)$$

$$A'(x) = -y_2 \cdot R/W = -\sin(x) \cdot \sec(x)/1 = -\sin(x)/\cos(x) = -\tan(x)$$

$$B'(x) = y_1 \cdot R/W = \cos(x) \cdot \sec(x)/1 = \cos(x)/\cos(x) = 1$$

Step 4: Integrate

$$A(x) = \int -\tan(x) dx = \ln|\cos(x)| \quad (\text{since } \int \tan(x) dx = -\ln|\cos(x)|)$$

$$B(x) = \int 1 \cdot dx = x$$

Step 5: PI

$$\text{PI} = A(x) \cdot y_1 + B(x) \cdot y_2 = \ln|\cos(x)| \cdot \cos(x) + x \cdot \sin(x)$$

$$y = C_1 \cdot \cos(x) + C_2 \cdot \sin(x) + \cos(x) \cdot \ln|\cos(x)| + x \cdot \sin(x)$$

● Example 16: $f(x) = e^x \cdot x$ (Exponential $\times$ Polynomial)

Problem: Solve $(D^2 - 2D + 1)y = e^x \cdot x$

CF

AE: $m^2 - 2m + 1 = 0 \rightarrow (m-1)^2 = 0 \rightarrow m=1$ (double). CF = $(C_1 + C_2x) \cdot e^x$

PI using Exponential Shift

$$PI = [1/(D^2 - 2D + 1)] \cdot e^x \cdot x = [1/(D-1)^2] \cdot e^x \cdot x$$

$$\text{Shift: } PI = e^x \cdot [1/(D+1-1)^2] \cdot x = e^x \cdot [1/D^2] \cdot x = e^x \cdot \int \int x \, dx \, dx$$

$$\int x \, dx = x^2/2$$

$$\int (x^2/2) \, dx = x^3/6$$

$$\therefore PI = e^x \cdot x^3/6$$

$$y = (C_1 + C_2x) \cdot e^x + (x^3/6) \cdot e^x$$

● Example 17: Wronskian Calculation – Three Functions

Problem: Find Wronskian of $y_1 = e^x$, $y_2 = e^{(2x)}$, $y_3 = e^{(3x)}$

Set up 3×3 determinant

$$W = \begin{vmatrix} e^x & e^{(2x)} & e^{(3x)} \\ e^x & 2e^{(2x)} & 3e^{(3x)} \\ e^x & 4e^{(2x)} & 9e^{(3x)} \end{vmatrix}$$

Factor out e^x , $e^{(2x)}$, $e^{(3x)}$ from columns

$$W = e^x \cdot e^{(2x)} \cdot e^{(3x)} \cdot \begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 4 & 9 \end{vmatrix}$$

$$= e^{(6x)} \cdot [1(18-12) - 1(9-3) + 1(4-2)]$$

$$= e^{(6x)} \cdot [6 - 6 + 2] = 2e^{(6x)} \neq 0$$

$$W = 2e^{(6x)} \neq 0 \rightarrow \text{Linearly Independent } \checkmark$$

● Example 18: Simultaneous ODE System

Problem: Solve $dx/dt - 2y = e^t$ and $dy/dt + 2x = 0$

D-notation ($D = d/dt$)

$$(D)x - 2y = e^t \quad \dots(1)$$

$$2x + (D)y = 0 \quad \dots(2)$$

Eliminate y: Operate (1) by D, multiply (2) by 2, add

$$D^2x - 2Dy = D \cdot e^t = e^t \quad \dots(1D)$$

$$4x + 2Dy = 0 \quad \dots(2 \times 2)$$

$$\text{Add: } D^2x + 4x = e^t \rightarrow (D^2 + 4)x = e^t$$

Solve for x

$$\text{AE: } m^2 + 4 = 0 \rightarrow m = \pm 2i. \text{ CF_}x = C_1 \cos(2t) + C_2 \sin(2t)$$

$$\text{PI_}x = e^t / (1 + 4) = e^t / 5$$

$$x = C_1 \cos(2t) + C_2 \sin(2t) + e^t / 5$$

Find y from equation (2): $y = -(1/2) \cdot dx/dt$

$$dx/dt = -2C_1 \sin(2t) + 2C_2 \cos(2t) + e^t / 5$$

$$y = -(1/2) \cdot [-2C_1 \sin(2t) + 2C_2 \cos(2t) + e^t / 5]$$

$$y = C_1 \sin(2t) - C_2 \cos(2t) - e^t / 10$$

$$\mathbf{x = C_1 \cos(2t) + C_2 \sin(2t) + e^t / 5 \ ; \ y = C_1 \sin(2t) - C_2 \cos(2t) - e^t / 10}$$

● Example 19: LCR Circuit Problem

Problem: An LCR circuit has $L = 1$ H, $R = 4 \Omega$, $C = 1/5$ F and EMF $E = 10\sin(t)$ V. Find the charge $q(t)$ given $q(0) = 0$, $q'(0) = 0$.

Step 1: Set up ODE

$$L \cdot q'' + R \cdot q' + (1/C) \cdot q = E(t)$$

$$1 \cdot q'' + 4 \cdot q' + 5 \cdot q = 10\sin(t)$$

$$(D^2 + 4D + 5)q = 10\sin(t)$$

Step 2: CF

$$AE: m^2 + 4m + 5 = 0 \rightarrow m = [-4 \pm \sqrt{(16-20)}]/2 = [-4 \pm 2i]/2 = -2 \pm i$$

$$\alpha = -2, \beta = 1. CF = e^{(-2t)} \cdot [C_1 \cos(t) + C_2 \sin(t)]$$

Step 3: PI for $10\sin(t)$

$$PI = [1/(D^2 + 4D + 5)] \cdot 10\sin(t)$$

$$\text{Replace } D^2 \rightarrow -1: PI = 10\sin(t)/(-1 + 4D + 5) = 10\sin(t)/(4 + 4D)$$

$$= 10\sin(t)/[4(1 + D)]$$

$$\text{Rationalize: } (5/2) \cdot [(1 - D)/(1 - D^2)] \cdot \sin(t) \rightarrow \text{replace } D^2 \rightarrow -1 \text{ in denom: } 1 - (-1) = 2$$

$$PI = (5/2) \cdot (1 - D) \cdot \sin(t)/2 = (5/4) \cdot (\sin(t) - \cos(t))$$

Step 4: General Solution

$$q(t) = e^{(-2t)} \cdot [C_1 \cos(t) + C_2 \sin(t)] + (5/4) \cdot [\sin(t) - \cos(t)]$$

Step 5: Apply $q(0)=0$

$$0 = C_1 + (5/4)(0 - 1) = C_1 - 5/4 \rightarrow C_1 = 5/4$$

Step 6: Apply $q'(0)=0$

$$q' = e^{(-2t)} [(-2C_1 + C_2)\cos(t) + (-2C_2 - C_1)\sin(t)] + (5/4)[\cos(t) + \sin(t)]$$

$$\text{At } t=0: (-2C_1 + C_2) + 5/4 = 0 \rightarrow -5/2 + C_2 + 5/4 = 0 \rightarrow C_2 = 5/4$$

$$q(t) = e^{(-2t)} \cdot (5/4) \cdot [\cos(t) + \sin(t)] + (5/4) \cdot [\sin(t) - \cos(t)]$$

● Example 20: SHM – Forced Oscillation with Resonance

Problem: Solve $d^2x/dt^2 + 4x = 8\cos(2t)$ (forced oscillation at resonance frequency)

CF

$$\text{AE: } m^2 + 4 = 0 \rightarrow m = \pm 2i. \text{ CF} = C_1 \cos(2t) + C_2 \sin(2t)$$

PI attempt

Replace $D^2 \rightarrow -4$: $\cos(2t)/(-4+4) = \cos(2t)/0 \rightarrow \text{FAILURE (resonance!)}$

PI using failure rule

$$\text{PI} = \text{Re} \{ [1/(D^2+4)] \cdot 8e^{(2it)} \}$$

$$\text{Since failure: PI} = \text{Re} \{ 8 \cdot x \cdot e^{(2it)} / (2 \cdot 2i \cdot 1) \} = \text{Re} \{ 8 \cdot t \cdot e^{(2it)} / (4i) \}$$

$$= \text{Re} \{ 2t \cdot e^{(2it)} / i \} = \text{Re} \{ 2t \cdot e^{(2it)} \cdot (-i) \}$$

$$= \text{Re} \{ 2t \cdot (\cos 2t + i \sin 2t) \cdot (-i) \}$$

$$= \text{Re} \{ 2t \cdot (-i \cos 2t + \sin 2t) \}$$

$$= 2t \cdot \sin(2t)$$

$$\therefore \text{PI} = 2t \cdot \sin(2t)$$

Interpretation

The amplitude grows without bound ($2t$ factor) – this is **RESONANCE**, a critical engineering concern. The system absorbs energy continuously, leading to catastrophic failure.

$$\mathbf{x(t) = C_1 \cos(2t) + C_2 \sin(2t) + 2t \cdot \sin(2t)}$$

● Example 21: Triple Repeated Root

Problem: Solve $(D - 2)^3 y = 0$

AE

$$(m-2)^3 = 0 \rightarrow m = 2, 2, 2 \text{ (triple root)}$$

CF

$$CF = (C_1 + C_2x + C_3x^2) \cdot e^{(2x)}$$

$$y = (C_1 + C_2x + C_3x^2) \cdot e^{(2x)}$$

● Example 22: PI for $f(x) = x \cdot \sin(x)$

Problem: Find PI of $(D^2 + 1)y = x \cdot \sin(x)$

Method: Use imaginary part of $[1/(D^2+1)] \cdot x \cdot e^{(ix)}$

$e^{(ix)} = \cos(x) + i \cdot \sin(x)$, so $x \cdot \sin(x) = \text{Im}\{x \cdot e^{(ix)}\}$

Exponential shift

$$[1/(D^2+1)] \cdot x \cdot e^{(ix)} = e^{(ix)} \cdot [1/((D+i)^2+1)] \cdot x$$

$$= e^{(ix)} \cdot [1/(D^2+2iD+i^2+1)] \cdot x$$

$$= e^{(ix)} \cdot [1/(D^2+2iD-1+1)] \cdot x$$

$$= e^{(ix)} \cdot [1/(D^2+2iD)] \cdot x$$

$$= e^{(ix)} \cdot [1/(2iD(1+D/2i))] \cdot x$$

Binomial expand and apply

$$= e^{(ix)} \cdot [1/(2iD)] \cdot (1 - D/2i + \dots) \cdot x$$

$$(1 - D/2i) \cdot x = x - 1/(2i) = x + i/2$$

$$[1/D] \cdot (x + i/2) = x^2/2 + ix/2$$

$$[1/(2i)] \cdot (x^2/2 + ix/2) = x^2/(4i) + x/4 = -ix^2/4 + x/4$$

Multiply by $e^{(ix)} = \cos x + i \sin x$, take Im part

$$e^{(ix)} \cdot (-ix^2/4 + x/4) = (\cos x + i \sin x)(x/4 - ix^2/4)$$

$$= \cos x \cdot x/4 - i \cos x \cdot x^2/4 + i \sin x \cdot x/4 + \sin x \cdot x^2/4$$

$$\text{Im part} = -x^2 \cos x/4 + x \sin x/4$$

$$\text{PI} = -(x^2/4) \cos x + (x/4) \sin x$$

$$\text{PI} = (x/4) \cdot \sin(x) - (x^2/4) \cdot \cos(x)$$

● Example 23: Variation of Parameters with $\tan(x)$

Problem: Solve $d^2y/dx^2 + y = \tan(x)$

CF

$$m^2 + 1 = 0 \rightarrow m = \pm i. \quad y_1 = \cos(x), \quad y_2 = \sin(x). \quad CF = C_1 \cos x + C_2 \sin x$$

$$W = \cos^2 x + \sin^2 x = 1$$

$$A'(x) = -\sin x \cdot \tan x = -\sin^2 x / \cos x = (\cos^2 x - 1) / \cos x = \cos x - \sec x$$

$$A(x) = \int (\cos x - \sec x) dx = \sin x - \ln|\sec x + \tan x|$$

$$B'(x) = \cos x \cdot \tan x = \sin x$$

$$B(x) = \int \sin x \, dx = -\cos x$$

$$PI = A \cdot \cos x + B \cdot \sin x$$

$$= [\sin x - \ln|\sec x + \tan x|] \cdot \cos x + [-\cos x] \cdot \sin x$$

$$= \sin x \cdot \cos x - \cos x \cdot \ln|\sec x + \tan x| - \cos x \cdot \sin x$$

$$= -\cos x \cdot \ln|\sec x + \tan x|$$

$$y = C_1 \cos x + C_2 \sin x - \cos x \cdot \ln|\sec x + \tan x|$$

● Example 24: $f(x) = e^x + \sin(x)$ – Sum of Two Functions

Problem: Solve $(D^2 - 1)y = e^x + \sin(x)$

CF

$$\text{AE: } m^2 - 1 = 0 \rightarrow (m-1)(m+1) = 0 \rightarrow m = 1, -1. \text{ CF} = C_1 e^x + C_2 e^{-x}$$

PI₁ for e^x : $f(1) = 1 - 1 = 0 \rightarrow \text{failure}$

$$f(D) = 2D \rightarrow f(1) = 2 \neq 0$$

$$\text{PI}_1 = x \cdot e^x / 2$$

PI₂ for $\sin(x)$: replace $D^2 \rightarrow -1$

$$\text{PI}_2 = \sin(x) / (-1 - 1) = \sin(x) / (-2) = -\sin(x) / 2$$

Total PI = PI₁ + PI₂

$$\text{PI} = (x/2)e^x - (1/2)\sin(x)$$

$$y = C_1 e^x + C_2 e^{-x} + (x/2)e^x - (1/2)\sin(x)$$

● Example 25: SHM – Finding Period and Amplitude

Problem: A particle executes SHM with $d^2x/dt^2 + 16x = 0$, $x(0)=3$, $x'(0)=8$. Find $x(t)$, amplitude, period.

Solution

$$\text{AE: } m^2 + 16 = 0 \rightarrow m = \pm 4i. \quad x(t) = C_1 \cos(4t) + C_2 \sin(4t)$$

ICs

$$x(0) = C_1 = 3. \quad x'(0) = 4C_2 = 8 \rightarrow C_2 = 2$$

Amplitude

$$A = \sqrt{C_1^2 + C_2^2} = \sqrt{9 + 4} = \sqrt{13}$$

Period

$$T = 2\pi/\omega = 2\pi/4 = \pi/2$$

$$x(t) = 3\cos(4t) + 2\sin(4t), \text{ Amplitude} = \sqrt{13}, \text{ Period} = \pi/2$$

● Example 26: PI for $f(x) = x^2 \cdot e^{(2x)}$

Problem: Find PI for $(D-2)^3 y = e^{(2x)} \cdot x^2$

PI

$$PI = [1/(D-2)^3] \cdot e^{(2x)} \cdot x^2$$

$$\text{Exponential shift: } PI = e^{(2x)} \cdot [1/(D+2-2)^3] \cdot x^2 = e^{(2x)} \cdot [1/D^3] \cdot x^2$$

$$[1/D^3] \cdot x^2 = \iiint x^2 \, dx \, dx \, dx$$

$$\text{First: } \int x^2 \, dx = x^3/3$$

$$\text{Second: } \int (x^3/3) \, dx = x^4/12$$

$$\text{Third: } \int (x^4/12) \, dx = x^5/60$$

$$PI = e^{(2x)} \cdot x^5/60$$

$$PI = (x^5/60) \cdot e^{(2x)}$$

● Example 27: Complete Solution with Mixed Forcing

Problem: Solve $(D^2 + 4D + 4)y = e^{(-2x)} + x^2$

CF

$$AE: (m+2)^2=0 \rightarrow m=-2 \text{ (double)}. CF=(C_1+C_2x)e^{(-2x)}$$

PI₁ for $e^{(-2x)}$

$$f(-2) = 4 - 8 + 4 = 0 \rightarrow \text{failure. } f(D) = 2D + 4 \rightarrow f(-2) = -4 + 4 = 0 \rightarrow \text{failure again!}$$

$$f'(D) = 2 \rightarrow f'(-2) = 2 \neq 0$$

$$PI_1 = x^2 \cdot e^{(-2x)} / f'(-2) = x^2 \cdot e^{(-2x)} / 2$$

PI₂ for x^2

$$PI_2 = [1/(D+2)^2] \cdot x^2 = [1/(4(1+D/2)^2)] \cdot x^2$$

$$= (1/4) \cdot (1+D/2)^{-2} \cdot x^2$$

$$= (1/4) \cdot (1 - D + 3D^2/4 - \dots) \cdot x^2$$

$$D(x^2) = 2x, D^2(x^2) = 2$$

$$= (1/4)(x^2 - 2x + 3 \cdot 2/4) = (1/4)(x^2 - 2x + 3/2)$$

$$y = (C_1 + C_2x)e^{(-2x)} + (x^2/2)e^{(-2x)} + (1/4)(x^2 - 2x + 3/2)$$

● **Example 28:** $(D^2 + a^2)y = \sec(ax)$ by Variation of Parameters

Problem: Solve $(D^2 + a^2)y = \sec(ax)$

CF

$$y_1 = \cos(ax), y_2 = \sin(ax). W = a$$

A'(x)

$$A' = -\sin(ax) \cdot \sec(ax)/a = -\tan(ax)/a \rightarrow A = \ln|\cos(ax)|/a^2$$

B'(x)

$$B' = \cos(ax) \cdot \sec(ax)/a = 1/a \rightarrow B = x/a$$

PI

$$PI = \cos(ax) \cdot \ln|\cos(ax)|/a^2 + x \cdot \sin(ax)/a$$

$$y = C_1 \cos(ax) + C_2 \sin(ax) + (x/a) \sin(ax) + (1/a^2) \cos(ax) \cdot \ln|\cos(ax)|$$

● Example 29: LCR with DC EMF (Step Response)

Problem: $L=2H$, $R=8\Omega$, $C=0.1F$, $E=100V$ (DC). Find $q(t)$ with $q(0)=0$, $i(0)=0$.

ODE

$$2q'' + 8q' + 10q = 100 \rightarrow (D^2 + 4D + 5)q = 50$$

CF

$$m^2 + 4m + 5 = 0 \rightarrow m = -2 \pm i. \text{ CF} = e^{(-2t)}(C_1 \cos t + C_2 \sin t)$$

PI for constant 50

$$PI = 50/f(0) = 50/5 = 10 \text{ (set } D=0)$$

Apply ICs $q(0)=0$

$$0 = C_1 + 10 \rightarrow C_1 = -10$$

Apply $i(0)=q'(0)=0$

$$q' = e^{(-2t)}[(-2C_1 + C_2)\cos t - (2C_2 + C_1)\sin t]$$

$$q'(0) = -2C_1 + C_2 = 20 + C_2 = 0 \rightarrow C_2 = -20$$

$$q(t) = 10 - e^{(-2t)}[10\cos(t) + 20\sin(t)] \text{ coulombs}$$

● Example 30: Damped Oscillation (Underdamped)

Problem: $d^2x/dt^2 + 2 \cdot dx/dt + 5x = 0$, $x(0)=0$, $x'(0)=4$

AE

$$m^2 + 2m + 5 = 0 \rightarrow m = -1 \pm 2i. \text{ CF} = e^{(-t)}(C_1 \cos 2t + C_2 \sin 2t)$$

Apply $x(0)=0$

$$C_1 = 0$$

$$x' = e^{(-t)}[(-C_1 + 2C_2)\cos 2t + (-C_2 - 2C_1)\sin 2t]; x'(0) = -C_1 + 2C_2 = 2C_2 = 4 \rightarrow C_2 = 2$$

Physical interpretation

System oscillates with decreasing amplitude (underdamped). The envelope of oscillation is $\pm 2e^{(-t)}$.

$$x(t) = 2e^{(-t)} \cdot \sin(2t) \text{ [Underdamped Oscillation]}$$

● Example 31: Fourth-Order ODE

Problem: Solve $(D^4 - 1)y = 0$

AE

$$m^4 - 1 = 0 \rightarrow (m^2 - 1)(m^2 + 1) = 0$$

$$m^2 = 1 \rightarrow m = \pm 1 \text{ (real distinct)}$$

$$m^2 = -1 \rightarrow m = \pm i \text{ (complex, } \alpha=0, \beta=1)$$

$$y = C_1 e^x + C_2 e^{-x} + C_3 \cos x + C_4 \sin x$$

● Example 32: PI for $f(x) = \cosh(x) = (e^x + e^{-x})/2$

Problem: Find PI for $(D^2 - 4)y = \cosh(x)$

Write $\cosh(x)$ as exponentials

$$\cosh(x) = (e^x + e^{-x})/2$$

$$PI_1 = [1/(D^2 - 4)] \cdot (e^x/2)$$

$$\text{Replace } D \rightarrow 1: 1/(1 - 4) \cdot (1/2) = -1/6$$

$$PI_2 = [1/(D^2 - 4)] \cdot (e^{-x}/2)$$

$$\text{Replace } D \rightarrow -1: 1/(1 - 4) \cdot (1/2) = -1/6$$

$$\text{Total PI} = -1/6 - 1/6 = -1/3$$

$$y = C_1 e^{2x} + C_2 e^{-2x} - 1/3$$

● Example 33: $(D^2 + D + 1)y = x^2 + 2x$

CF

$$m^2 + m + 1 = 0 \rightarrow m = (-1 \pm \sqrt{-3})/2 = -1/2 \pm (\sqrt{3}/2)i$$

$$CF = e^{(-x/2)} \cdot [C_1 \cos(\sqrt{3}x/2) + C_2 \sin(\sqrt{3}x/2)]$$

PI

$$PI = [1/(1+D+D^2)] \cdot (x^2+2x)$$

$$= (1+D+D^2)^{-1} \cdot (x^2+2x)$$

$$= [1 - (D+D^2) + (D+D^2)^2 - \dots] \cdot (x^2+2x)$$

Keep terms $\leq D^2$:

$$\approx (1 - D - D^2 + D^2) \cdot (x^2+2x) \quad [D^2 \cdot D \text{ terms vanish on } x^2]$$

$$= (1 - D) \cdot (x^2+2x)$$

$$D(x^2+2x) = 2x+2$$

$$PI = (x^2+2x) - (2x+2) = x^2 - 2$$

$$y = e^{(-x/2)} [C_1 \cos(\sqrt{3}x/2) + C_2 \sin(\sqrt{3}x/2)] + x^2 - 2$$

● Example 34: Variation of Parameters for e^x/x

Problem: Solve $(D^2 - 2D + 1)y = e^x/x^2$

CF

$$(m-1)^2 = 0 \rightarrow m=1 \text{ (double)}. y_1 = e^x, y_2 = xe^x. CF = (C_1 + C_2x)e^x$$

Wronskian

$$W = y_1 y_2' - y_2 y_1' = e^x(e^x + xe^x) - xe^x \cdot e^x = e^{2x}$$

$$R = e^x/x^2$$

$$A' = -xe^x \cdot (e^x/x^2)/e^{2x} = -xe^{2x}/(x^2 \cdot e^{2x}) = -1/x$$

$$A = -\ln|x|$$

$$B' = e^x \cdot (e^x/x^2)/e^{2x} = 1/x^2$$

$$B = -1/x$$

PI

$$PI = -\ln|x| \cdot e^x + (-1/x) \cdot xe^x = e^x(-\ln|x| - 1)$$

$$y = (C_1 + C_2x)e^x - e^x(1 + \ln|x|)$$

● Example 35: Overdamped LCR

Problem: $L=1\text{H}$, $R=5\Omega$, $C=1/6\text{F}$, $E=0$. $q(0)=Q_0$, $i(0)=0$. Find $q(t)$.

ODE

$$q''+5q'+6q=0. \text{ AE: } m^2+5m+6=0 \rightarrow (m+2)(m+3)=0 \rightarrow m=-2, -3$$

CF

$$q=C_1e^{(-2t)}+C_2e^{(-3t)}$$

$$q(0)=Q_0$$

$$C_1+C_2=Q_0$$

$$i(0)=q'(0)=0$$

$$-2C_1-3C_2=0 \rightarrow C_1=-3C_2/2. \text{ Substituting: } -3C_2/2+C_2=Q_0 \rightarrow C_2=-2Q_0. C_1=3Q_0$$

Nature

Overdamped (two distinct real negative roots). Charge decays without oscillation.

$$q(t) = 3Q_0e^{(-2t)} - 2Q_0e^{(-3t)} \text{ [Overdamped Decay]}$$

● Example 36–40: Quick but Complete Examples

Ex.36: Solve $(D^3-D)y = 2e^x+3$. AE: $m^3-m=0 \rightarrow m(m^2-1)=0 \rightarrow m=0, 1, -1$. CF= $C_1+C_2e^x+C_3e^{-x}$.

PI for $2e^x$: $f(1)=0$ (failure), $PI_1=2 \cdot x \cdot e^x/f(1)=2xe^x/(3-1)=xe^x$

PI for 3 (constant): $PI_2=3/f(0) \rightarrow f(0)=0$ (failure). Use $[1/(D(D^2-1))]\cdot 3$, treat as $[1/D] \cdot [1/(D^2-1)] \cdot 3 = [1/D] \cdot (3/(0-1)) = [1/D] \cdot (-3) = -3x$

Answer: $y=C_1+C_2e^x+C_3e^{-x}+xe^x-3x$

Ex.37: Show $y_1=x$, $y_2=x^2$ are linearly independent. $W=|x \ x^2; 1 \ 2x|=2x^2-x^2=x^2 \neq 0$ for $x \neq 0$. ✓

Ex.38: Solve $(D^2+4D+13)y=e^{(-2x)}\sin(3x)$. CF: $m=-2 \pm 3i$, CF= $e^{(-2x)}(C_1\cos 3x+C_2\sin 3x)$. PI: shift by -2 , get $[1/(D^2+9)] \cdot \sin 3x$ (failure). PI= $-x\cos(3x)/6 \cdot e^{(-2x)}$. Answer: $y=e^{(-2x)}[(C_1-x/6)\cos 3x+C_2\sin 3x]$

Ex.39: $(D^2-4)y=4x-2$. CF= $C_1e^{(2x)}+C_2e^{(-2x)}$. PI= $[1/(D^2-4)](4x-2)=[-1/4(1-D^2/4)](4x-2) \approx -(1/4)(4x-2)-(1/4)(0)=-x+1/2$. Ans: $y=C_1e^{(2x)}+C_2e^{(-2x)}-x+1/2$

Ex.40: Mass-spring: $d^2x/dt^2+9x=\cos(2t)$, $x(0)=1, x'(0)=0$. CF= $C_1\cos 3t+C_2\sin 3t$. PI= $\cos(2t)/(9-4)=\cos(2t)/5$. $x(0)=C_1+1/5=1 \rightarrow C_1=4/5$. $x'(0)=3C_2=0 \rightarrow C_2=0$. **Ans:** $x=4\cos(3t)/5+\cos(2t)/5$

Part A: 2-Mark Questions (Definitions & Short Answers)

1. Define the order and degree of a differential equation. Find the order and degree of: $(d^2y/dx^2)^3 + (dy/dx)^4 = x$
2. What is meant by a linear differential equation? Give two examples each of linear and non-linear ODEs.
3. Distinguish between homogeneous and non-homogeneous linear ODEs with examples.
4. Define the Complementary Function (CF). What is its relation to the Auxiliary Equation?
5. Define the Particular Integral (PI). Why is PI needed in addition to CF?
6. State the general solution structure for a non-homogeneous linear ODE.
7. Write the CF for: (a) $m=2,3$ (b) $m=-1,-1$ (c) $m=1\pm 2i$ (d) $m=\pm 3i$
8. Define the Wronskian. What does $W=0$ vs $W\neq 0$ tell us about solutions?
9. State the standard PI formula for $f(x) = e^{ax}$. When does the failure case occur?
10. State the standard PI formula for $f(x) = \sin(ax)$. What replacement is used for D^2 ?
11. Write the governing ODE for an LCR series circuit. Identify which term corresponds to each component.
12. Write the standard SHM equation and state the meaning of ω (angular frequency), period T , and amplitude A .
13. Form the Auxiliary Equation for: $d^3y/dx^3 + 2d^2y/dx^2 - 5dy/dx - 6y = 0$
14. Solve the AE: $m^2 - 7m + 12 = 0$. State the case and write the CF.
15. When does resonance occur in a forced oscillation? What is its practical significance?

Part B: 5-Mark Questions (Full Solutions Required)

16. Solve: $d^2y/dx^2 + 6dy/dx + 9y = 0$
17. Solve: $d^2y/dx^2 - 2dy/dx + 5y = 0$
18. Solve: $d^3y/dx^3 - 7dy/dx - 6y = 0$
19. Solve: $(D^4 + 8D^2 + 16)y = 0$
20. Solve: $(D^2 + 4D + 4)y = e^{-2x}$
21. Solve: $(D^2 + 1)y = \sin(x) \cdot \cos(x)$ [Hint: use product-to-sum identity]
22. Solve: $(D^2 - 4)y = x^2$
23. Solve: $(D^2 + 2D + 1)y = x \cdot e^{-x}$
24. Solve: $(D^2 + 4)y = x \cdot \sin(2x)$ [resonance-type PI]
25. Solve: $(D^2 - D - 2)y = e^{2x} + 3\cos(x)$
26. Solve using Variation of Parameters: $d^2y/dx^2 + y = \operatorname{cosec}(x)$
27. Solve using Variation of Parameters: $d^2y/dx^2 - 2dy/dx + y = e^x \cdot \ln(x)$
28. Find Wronskian of $y_1 = \sin(2x)$, $y_2 = \cos(2x)$ and test linear independence.

29. Solve the simultaneous system: $Dx + 2y = e^t$; $2x + Dy = 0$ ($D = d/dt$)
30. Solve the simultaneous system: $Dx - y = t$; $x + Dy = 1$
31. Solve: $(D^2 + 3D + 2)y = e^x \cdot \sin(x)$
32. Solve: $(D^2 - 1)y = x^2 \cdot e^x$
33. Solve: $d^2y/dx^2 + 4y = 4\tan(2x)$ using Variation of Parameters
34. LCR Circuit: $L=1H$, $R=2\Omega$, $C=1/2F$, $E=e^{(-t)} \sin(t)$. Find general charge $q(t)$.
35. A mass-spring system satisfies $m \cdot x'' + kx = F_0 \cdot \cos(\omega t)$ where $\omega^2=k/m$. Solve and describe resonance.

Part C: Application Problems (10 Problems)

36. A spring stretches 0.1m under 10N. A mass of 2kg is attached. Set up and solve the SHM equation for $x(0)=0.2m$, $x'(0)=0$.
37. An LCR circuit has $L=0.5H$, $C=0.02F$, $R=0$ (no resistance). Find the natural frequency. If $E=100\cos(10t)$, solve for charge.
38. For overdamped LCR ($R^2 > 4L/C$), prove charge decays without oscillation. Find $q(t)$ with $q(0)=Q$, $q'(0)=0$.
39. Critically damped system: $d^2x/dt^2 + 6dx/dt + 9x = 0$, $x(0)=0$, $x'(0)=3$. Solve and describe the motion.
40. Two tanks with interconnecting pipes lead to the system: $dx/dt = -2x + y$; $dy/dt = x - 2y$. Solve for $x(0)=10$, $y(0)=0$.
41. Pendulum: for small angles, $d^2\theta/dt^2 + (g/L)\theta = 0$. Find $\theta(t)$ if $\theta(0)=\theta_0$, $\theta'(0)=0$. Find period.
42. Prove that if y_1 and y_2 are solutions of $(D^2 + P \cdot D + Q)y = 0$, then so is $c_1y_1 + c_2y_2$ (Principle of Superposition).
43. Damped SHM: $d^2x/dt^2 + 4dx/dt + 20x = 0$. Identify the type of damping and find $x(t)$ with $x(0)=1$, $x'(0)=0$.
44. Find the PI of $d^2y/dx^2 + 4y = 8x^2 - 4x$ using the operator method with binomial expansion. Show all steps.
45. Solve the coupled system: $d^2x/dt^2 = y$, $d^2y/dt^2 = x$, with $x(0)=1$, $x'(0)=0$, $y(0)=-1$, $y'(0)=0$.

16.1 Key Theorems

Theorem 1 – Superposition Principle:

If y_1 and y_2 are any two solutions of the linear homogeneous ODE $L[y]=0$, then $c_1y_1+c_2y_2$ is also a solution, for any constants c_1, c_2 .

Proof sketch: $L[c_1y_1+c_2y_2] = c_1L[y_1]+c_2L[y_2] = c_1 \cdot 0 + c_2 \cdot 0 = 0$. (L is linear operator)

Theorem 2 – Existence and Uniqueness:

For the n -th order linear ODE with continuous coefficients and n initial conditions at a point x_0 , there exists exactly one solution defined in a neighborhood of x_0 .

Theorem 3 – Structure of General Solution:

The general solution of $L[y]=f(x)$ is $y = y_c + y_p$ where y_c is the CF (n arbitrary constants) and y_p is any particular solution.

Theorem 4 – Abel's Identity for Wronskian:

For solutions $y_1, \dots, y_n$ of $y^{(n)} + P_1y^{(n-1)} + \dots + P_ny = 0$:

$$W(x) = W(x_0) \cdot \exp\left(-\int P_1 dx\right)$$

Hence W is either identically zero or never zero. This means solutions are either always dependent or always independent.

Theorem 5 – Exponential Shift:

For any polynomial operator $f(D)$ and any function $V(x)$:

$$f(D)[e^{ax} \cdot V(x)] = e^{ax} \cdot f(D+a)[V(x)]$$

And therefore: $[1/f(D)] \cdot [e^{ax} \cdot V(x)] = e^{ax} \cdot [1/f(D+a)] \cdot V(x)$

16.2 Master Formula Sheet

===== AUXILIARY EQUATION ===== Replace D^k by m^k in $f(D)=0$ to get AE. AE is a polynomial equation in m of degree n . =====

CF CASES ===== Case 1: $m_1 \neq m_2$ (distinct real) $\rightarrow C_1 e^{(m_1 x)} + C_2 e^{(m_2 x)}$ Case 2: $m_1 = m_2 = m$ (double real) $\rightarrow (C_1 + C_2 x) e^{(mx)}$ Case 2b: Triple $m \rightarrow (C_1 + C_2 x + C_3 x^2) e^{(mx)}$ Case 3: $m = \alpha + i\beta$ (complex pair) $\rightarrow e^{(\alpha x)} [C_1 \cos(\beta x) + C_2 \sin(\beta x)]$ Case 4: $m = \alpha + i\beta$ (repeated complex) $\rightarrow e^{(\alpha x)} [(C_1 + C_2 x) \cos(\beta x) + (C_3 + C_4 x) \sin(\beta x)]$ Case 5: $m = i\beta$ (pure imaginary) $\rightarrow C_1 \cos(\beta x) + C_2 \sin(\beta x)$ ===== PI FORMULAS =====

===== Type 1: $[1/f(D)] \cdot e^{(ax)} = e^{(ax)}/f(a)$, provided $f(a) \neq 0$ If $f(a)=0$: $x \cdot e^{(ax)}/f'(a)$ [multiply x , differentiate $f(D)$ once w.r.t. D] If $f(a)=f'(a)=0$: $x^2 \cdot e^{(ax)}/f''(a)$, etc. Type 2: $[1/f(D^2)] \cdot \sin(ax) = \sin(ax)/f(-a^2)$, provided $f(-a^2) \neq 0$ $[1/f(D^2)] \cdot \cos(ax) = \cos(ax)/f(-a^2)$, provided $f(-a^2) \neq 0$ If $f(-a^2)=0$: Failure $\rightarrow$ use $x \dots$ method Type 3: $[1/f(D)] \cdot x^n = \text{expand } f(D)^{-1} \text{ as power series in } D$, apply to x^n (Series terminates at $D^{(n+1)}$) Type 4: $[1/f(D)] \cdot e^{(ax)} \cdot V(x) = e^{(ax)} \cdot [1/f(D+a)] \cdot V(x)$ [Exponential Shift] Type 5: $[1/f(D)] \cdot x \cdot \sin(ax) = \text{Im}\{[1/f(D)] \cdot x \cdot e^{(iax)}\}$ [Use shift+binomial]

===== VARIATION OF PARAMETERS ===== For $y'' + Py' + Qy = R(x)$, CF = $C_1 y_1 + C_2 y_2$: PI = $y_2 \int (y_1 R/W) dx - y_1 \int (y_2 R/W) dx$ where $W = y_1 y_2' - y_2 y_1'$

===== LCR CIRCUIT ===== $L \cdot q'' + R \cdot q' + (1/C) \cdot q = E(t)$ Natural frequency: $\omega_0 = 1/\sqrt{LC}$ Damping factor: $\alpha = R/(2L)$ =====

SHM ===== $x'' + \omega^2 x = 0 \rightarrow x = A \cdot \cos(\omega t + \phi)$ Period: $T = 2\pi/\omega$ Amplitude: $A = \sqrt{C_1^2 + C_2^2}$ Frequency: $f = \omega/(2\pi)$ Forced SHM: $x'' + \omega^2 x = F \cdot \cos(pt)$ • $p \neq \omega$: Beats/regular forced oscillation • $p = \omega$: RESONANCE (amplitude grows as $t \cdot \sin(\omega t)$)

16.3 PI Method Selection Guide

f(x) Type	Identify By	Method to Use	Key Formula
Pure exponential	$f(x)=e^{(ax)}$	Replace $D \rightarrow a$	$e^{(ax)}/f(a)$
Sine/Cosine	$\sin(bx)$ or $\cos(bx)$	Replace $D^2 \rightarrow -b^2$	$\sin(bx)/f(-b^2)$
Polynomial	x^n	Binomial expansion of $1/f(D)$	Terminate at $D^{(n+1)}$
Exp $\times$ trig	$e^{(ax)} \sin(bx)$	Exponential shift then Type 2	Shift $D \rightarrow D+a$
Exp $\times$ poly	$e^{(ax)} x^n$	Exponential shift then Type 3	Shift $D \rightarrow D+a$
$x \cdot \sin/\cos$	$x \cdot \sin(bx)$	Im/Re of exponential method	Complex exponential
sec, tan, ln, $1/x$	Cannot be expanded	Variation of Parameters	Use Wronskian formula

16.4 Quick Revision Notes

⚡ MEMORY TRICKS FOR EXAMS

1. **DARE** to solve: **D**efine type $\rightarrow$ **A**uxiliary equation $\rightarrow$ **R**oots $\rightarrow$ CF/PI $\rightarrow$ **E**xpress solution

2. AE Roots – "RIDE" the cases:

R = Real distinct: $e^{(m_1x)} + e^{(m_2x)}$

I = Imaginary (complex): $e^{(\alpha x)} \cdot [\cos + \sin]$

D = Double/Repeated: $(C_1 + C_2x) \cdot e^{(mx)}$

E = Equal imaginary (repeated complex): $(C_1 + C_2x)\cos + (C_3 + C_4x)\sin$

3. For PI: "ESPA" mnemonic:

E = Exponential (replace $D \rightarrow a$)

S = Sine/Cosine (replace $D^2 \rightarrow -a^2$)

P = Polynomial (binomial expand)

A = All others (Variation of Parameters)

4. **FAILURE** always means: multiply by x (repeat if needed)

5. Always verify: number of arbitrary constants = order of ODE

✗ MOST COMMON EXAM MISTAKES – AVOID THESE!

1. Forgetting to check the failure condition for PI (checking if $f(a)=0$ for exponentials)
2. Writing $C_1 \cdot e^{(mx)} + C_2 \cdot e^{(mx)}$ for a double root – WRONG! Use $(C_1 + C_2x) \cdot e^{(mx)}$
3. Replacing D alone (not D^2) when using sin/cos method
4. Forgetting that $1/D$ means integration ($\int \dots dx$)
5. Not verifying that constants in simultaneous ODE solutions are related (not all independent)
6. Forgetting to add PI to CF in the final answer
7. Confusing ω (angular frequency) with f (frequency): $\omega = 2\pi f$
8. In LCR, not converting from q -form to i -form correctly when required
9. For variation of parameters, forgetting to divide by W (Wronskian) in the integrals
10. Expanding binomial incorrectly – always factor out the constant term first

Step 1: Always write the ODE in D-operator form first.

Step 2: Identify: homogeneous (RHS=0) or non-homogeneous (RHS≠0)

Step 3: Find AE, solve it, note all root types

Step 4: Write CF using the root-type table (memorize the 5 cases)

Step 5: For PI, classify $f(x)$ and choose the right method

Step 6: Always check for failure before applying standard PI formula

Step 7: Add CF+PI for the general solution

Step 8: Apply initial/boundary conditions if given

Step 9: Box your final answer clearly

Time estimate: Simple CF/PI → 5 min. Full IVP → 10–15 min. Variation of parameters → 15–20 min.

16.5 Summary Table – All CF Forms at a Glance

Root of AE	Example	CF Term	Number of Constants
Single real m	$m=3$	$C \cdot e^{(3x)}$	1
Two distinct real m_1, m_2	$m=2, 5$	$C_1 e^{(2x)} + C_2 e^{(5x)}$	2
Double real m (repeated)	$m=-1, -1$	$(C_1 + C_2 x) e^{(-x)}$	2
Triple real m	$m=2, 2, 2$	$(C_1 + C_2 x + C_3 x^2) e^{(2x)}$	3
Complex pair $\alpha \pm i\beta$	$m=1 \pm 3i$	$e^x (C_1 \cos 3x + C_2 \sin 3x)$	2
Pure imaginary $\pm i\beta$	$m=\pm 2i$	$C_1 \cos(2x) + C_2 \sin(2x)$	2
Complex pair repeated	$m=\pm 2i$ (twice)	$(C_1 + C_2 x) \cos 2x + (C_3 + C_4 x) \sin 2x$	4
Mix: real + complex	$m=1, m=0 \pm 2i$	$C_1 e^x + C_2 \cos 2x + C_3 \sin 2x$	3

16.6 Glossary of Key Terms

Term	Meaning
ODE	Ordinary Differential Equation – involves derivatives w.r.t. ONE variable
Order	Highest derivative present in the ODE
Degree	Power of the highest derivative (after rationalizing)
Linear ODE	y and all its derivatives appear to power 1 only, no products of derivatives
Homogeneous	RHS = 0

Non-Homogeneous	$\text{RHS} = f(x) \neq 0$
Auxiliary Equation	Polynomial in m obtained by substituting $y=e^{mx}$
CF	Complementary Function – general solution of homogeneous part
PI	Particular Integral – one specific solution of non-homogeneous ODE
Operator D	$D=d/dx$; allows algebraic manipulation of ODE
Wronskian	Determinant used to test linear independence of solutions
Resonance	Forcing frequency = natural frequency $\rightarrow$ amplitude grows without bound
Overdamped	$R^2 > 4L/C$ (or equivalent) $\rightarrow$ two distinct real negative roots, no oscillation
Underdamped	$R^2 < 4L/C \rightarrow$ complex roots with negative real part, decaying oscillation
Critically Damped	$R^2 = 4L/C \rightarrow$ equal real roots, fastest decay without oscillation

IMPORTANT NOTE FOR STUDENTS

This module covers the entire Unit II syllabus for B.Tech Engineering Mathematics. Every topic – from basic definitions to complex variation of parameters and circuit applications – has been explained from absolute zero with no steps skipped.

Recommended Study Approach:

Day 1: Sections 1–5 (Definitions, CF, all root cases) + Examples 1–10

Day 2: Sections 6–8 (PI methods) + Examples 11–20

Day 3: Sections 9–11 (Wronskian, VoP, Simultaneous) + Examples 21–30

Day 4: Sections 12–13 (Applications) + Examples 31–40

Day 5: Section 16 (Formula Sheet + Revision) + All exercises

For exam day: Focus on Section 16 (Formula Sheet) and the Master CF Table. Practice writing CF for each case from memory in under 2 minutes.